MILITARY SPECIFICATION

BALLISTICS COMPUTER, SUBASSEMBLY: 10543800 (COMPUTER UNIT)

1. SCOPE

1.1 This specification covers one type of electro-mechanical assembly known as Ballistics Computer, Subassembly; 10543800 (Computer Unit).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U. S. Army, Frankford Arsenal

F10543800	Ballistics Computer, Subassembly: 10543800 (Computer Unit).

PACKAGING DATA SHEET

MIL-P-14232/ P10543800	Packaging for Computer Unit: 10543800

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed

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by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The Computer Unit shall be manufactured in accordance with Drawing F10543800 and drawings pertaining thereto.

3.2 General Specification.-The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

a. Order of precedence.
b. Dimensions and tolerances.
c. Inorganic protective surface finishes.
d. Part identification and marking.
e. Workmanship.

3.3 Environmental.

3.3.1 Storage Temperatures.- There shall be no evidence of physical failures such as glass breakage or casting failures due to storage temperatures ranging from -80° to +160°F.

3.3.2 Operating Temperatures.-The unit shall meet all the requirements of 3.5.2 through 3.5.15 inclusive, while exposed to and thermally stabilized at, ambient temperatures of -40° and +150°F.

3.3.3 Vibration.- The unit shall meet the requirements of 3.4.1 through 3.5.15 after being vibrated in a vertical plane for five minutes ±15 seconds at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second.

3.3.4 Water resistance (Sealing).-The interior of the unit shall show no evidence of water damage after being subjected to a simulated rainfall.

3.4 Electrical.

3.4.1 Continuity.-The continuity of the unit shall conform to Drawing F10543800.

3.5 Performance.

3.5.1 Input Power.- The unit shall meet the requirements of 3.5.2 through 3.5.15 when excited with 24VDC ± 1% between pins D and B of 1J23, pin D to be positive, and 115 volts RMS ±1%, 400 CPS ±1% between pins A and C of 1J23. Harmonic distortion shall not exceed 2%. Variation in harmonic distortion shall not exceed 0.5% during tests. Source impedance of the A.C. supply shall be less than 0.1 ohm.

3.5.2 Ammunition Selection. - Switch ISI shall step to and indicate selection of operator selected ammunition when 1J24 and 1J26 are connected

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to appropriate external switches and indicator devices. The maximum time required for 3 steps (ammunition 1 to 4, 2 to 1, etc.) shall not exceed 1 second.

3.5.3 Effective Full Charge Count.- The effective full charge dial shall advance 50 counts ±1 count for each 50 actuations of a test switch connected to IJ26 to simulate action of the EFC switch.

3.5.4 Power Supply Output.- With a 250 MA load connected across IJ26-J and D, the D.C. voltage at pin J shall be 24.0 D.C. ±5%. The ripple across the load shall have a maximum true R.M.S. value of 0.45 volt. The ripple frequency shall be twice that of the A.C. supply.

3.5.5 Elevation Amplifier Outputs.

3.5.5.1 Gunner's Elevation Channel.- The maximum input signal required at the gunner's elevation jump jack to drive the amplifier into saturation (maximum current flow through the center tap of a 160 ohm center tapped resistive load connected across pins BB and DD of IJ26, with the center tap supplied from pin CC of IJ26) shall not exceed 0.08 volts RMS with gain control R20 full counterclockwise. All other test jacks with the exception of the amplifier input, shall be grounded during the test.

3.5.5.2 Commander's Elevation Channel.- The maximum input signal required at the commander's elevation jump jacks to drive the amplifier into saturation (maximum current flow through the center top of a 160 ohm center tapped resistive load connected across pins i and j of IJ24, with the center tap supplied from pin CC of IJ26) shall not exceed 0.08 volts RMS with gain control R20 full counterclockwise. All other test jacks with the exception of the amplifier input shall be grounded during the test.

3.5.6.2 Commander's Deflection Channel.- The maximum input signal required at the commander's deflection jump jack to drive the amplifier into saturation (maximum current flow through the center tap of a 160 ohm center tapped resistive load connected across pins n amd p of IJ24 with the center tap supplied from pin CC of IJ26) shall not exceed 0.08 volts RMS with gain control R28 full counterclockwise. All other test jacks with the exception of the amplifier input shall be grounded during the test.

3.5.7 Range Servo Amplifier Output. - The maximum input signal required at the range servo follow-up jack to drive the amplifier into saturation (maximum current flow through the center tap of a 160 ohm center tapped resistive load connected across pins EE and FF of IJ25, with the center tap supplied from pin CC of IJ 26) shall not exceed 0.08 volts RMS with gain control R20 full counterdlockwise. All other test jacks with the exception of the amplifier input shall be grounded during the test.

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3.5.8 Parameter resistor package.-The resistances given in Table I shall be neasured across the terminals indicated.

TABLE I

RESISTANCE	TOLERANCE	FROM	TO	CONDITIONS
1 ohm max.	--	IJ24-a	Commander's "D" Parallax Jack	--
13,283 ohms	± 1%	IJ23-B	Commander's "E" Parallax Jack	Ground IJ25-D and Gunner's "D" amp input
23,730 ohms	± 1%	IJ23-B	Gunner's "D" Parallax Jack	Ground IJ25-D and Gunner's "D" amp input
3,552 ohms	± 1%	IJ23-B	Gunner's "E" Parallax Jack	Ground IJ25-D and Gunner's "E" amp input
16,886 ohms	± 1%	IJ23-B	Commander's "E" Drift Sin C Jack	Ground IJ25-F and Commander's "E" amp input
1,660 ohms	± 1%	IJ23-B	Commander's "E" Droop Sin C Jack	Ground IJ27-F and Commander's "D" amp input
77.8 ohms	± 1%	IJ23-B	Commander's "E" Droop Jack	Ground Commander's "E" amp input

3.5.9 Ammunition Resistor Package.- With switch ISI indexed to select M409, HEAT ammunition, the resistances of Table II shall be measured. A maximum additional resistance of 1.0 ohm shall be permitted for resistance of connection, leads, and harness wires.

TABLE II

RESISTANCE	TOLERANCE	FROM	TO
924.13 ohms	±0.1%	IJ25-B	IJ26-C
69,645 ohms	± 5% (Ground IJ25-A and Commander's "D" Drift Sin C Jack)	IJ25-F	IJ25-E
2581.9 ohms	± 1% (Ground Gunner's "D" Droop Sin C Jack)	IJ27-F	IJ27-E

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TABLE II (Continued)

RESISTANCE	TOLERANCE	FROM	TO
2581.9 ohms	± 1% (Ground Gunner's "D") (Droop Sin C Jack)	IJ27-F	IJ25-E
28.10 ohms	± 5%	IJ25-G	IJ27-F
473.13 ohms	± 0.1%	IJ25-M	IJ26-a
1 ohm max.		IJ25-G	IJ25-H
1 ohm max.		IJ25-P	Gunner's deflection lead jack
100,000 ohms	± 5% (Ground Gunner's "E") (Droop Jack)	IJ25-R	IJ25-E
213.35 ohms	±0.1%	IJ25-S	IJ27-P
1 ohm max.		IJ25-S	IJ25-T
1 ohm max.		IJ25-T	IJ25-U
1 ohm max.		IJ25-U	IJ25-V
1 ohm max.		IJ25-V	IJ25-W
2751.95 ohms	± .1%	IJ25-W	IJ25-X
3313.71 ohms	± .1%	IJ25-X	IJ25-Y
2541.88 ohms	± .1%	IJ25-Y	IN25-Z
1960.82 ohms	± .1%	IJ25-Z	IJ25-a
1538.95 ohms	± .1%	IJ25-a	IJ25-b
1247.63 ohms	± .1%	IJ25-b	IJ25-c
1029.50 ohms	± .1%	IJ25-c	IJ25-d
862.94 ohms	± .1%	IJ25-d	IJ25-e
732.69 ohms	± .1%	IJ25-e	IJ25-f
478.84 ohms	± .1%	IJ25-f	IJ25-G

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3.5.10 Digital to Analog Converter.

3.5.10.1 Relay Contacts.- An open circuit shall be exhibited between pins n, p, q, r, s, t, u, v, w, x, y of IJ25 and pin B of IJ 23 when relay K12 is de-energized.

3.5.10.2 Relay diodes.- Relay K12 shall be energized with a 24 VDC source. A 6 VDC source in series with 500 ohms shall be connected across pin IJ24-a and pins IJ25 -n,p,q,r,s,t,u,v,w,x, and y respectively. The positive terminal of the 6 VDC source shall be connected to IJ24-a. The voltage drop across the 500 ohm resister shall not be less than 5.0 VDC when the source is connected to the specified pins.

3.5.10.3 Converter Scale Factor Adjustment. - The voltage ratios of Table III shall be obtained for the specified ranges. The maximum ratio error due to instrumentation, with all relays de-energized, shall be .000200. The instrumentation ratio error measured shall be algebraically added to all voltage ratios in Table III.

TABLE III

RANGE	VOLTAGE RATIO	TOLERANCE
2000	.694444	± .002222
1000	.347222	± .001111
800	.277777	± .000889
400	.138888	± .000445
200	.069444	± .001389
100	.034722	± .000695
80	.027777	± .000548
40	.013888	± .000274
20	.006944	± .000208
10	.003472	± .000104
5	.001736	± .000052

3.5.11 Wind X sine cant.- The voltage ratios of Table IV shall be obtained when the specified inputs are applied.

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TABLE IV

CANT INPUTS	WIND INPUT	RATIO	TOLERANCE
6.0 Volts RMS Pi Ø	9.0 Volts RMS Pi Ø	.250000	± .0125
3.0 Volts RMS Pi Ø	9.0 Volts RMS Pi Ø	.125000	± .00625
0 Volts	9.0 Volts RMS Pi Ø	.000000	± .0010
0 Volts	9.0 Volts RMS Pi Ø	.000000	± .0010
3.0 Volts RMS Zero Ø	9.0 Volts RMS Pi Ø	.125000	± .00625
6.0 Volts RMS Zero Ø	9.0 Volts RMS Pi Ø	.250000	± .0125

3.5.12 Commander's Override Circuit.- Switch ISI shall be indexed to the ammunition no. 1 or M409 HEAT position. The A.C. and D.C. sources of 3.5.1 shall be removed. The resistances of Table V shall be measured.

TABLE V

ALL RELAYS DE-ENERGIZED

RESISTANCE	FROM	TO
500 ohm max.	IJ26-a	IJ25-M
2000 ohm max.	IJ26-Y	IJ23-B
1 ohm max.	IJ26-Y	IJ26-u
35 ohm max.	IJ27-F	IJ25-H
230 ohm max.	IJ27-P	IJ25-S
1 ohm max.	(Range Servo Range Input Jack)	IJ24-d
	K_1 ENERGIZED	
2000 ohm max.	IJ24-J	IJ23-B
500 ohm max.	IJ24-X	IJ25-M
1 ohm max.	IJ23-B	IJ24-U
	K_2 ENERGIZRD	
35 ohm max.	IJ23-B	IJ25-H
250 ohm max.	IJ23-B	IJ25-S

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TABLE V (Continued)

K_3 ENERGIZED

RESISTANCE	FROM	TO
Open	IJ25-M	IJ26-a

K_4 ENERGIZED

1 OHM MAX.	IJ25-HH	Range Servo Range Input Jack

3.5.13 Boresight-Norman Circuitry.- Switch ISI shall be indexed to the ammunition No. 1 or "HEAT M409" position. The A.C. and D.C. sources of 3.5.1 shall be removed.

3.5.13.1 Relay K_1. - A 1.0 volt RMS, 400 cycle signal shall be injected at the gunner's deflection channel ØS x sin C jack and the gunner's elevation channel ØS x cos C jack. The signal shall be present at the amplifier input jacks when K_1 is de-energized. The signal shall be removed from the input jacks when K_1 is energized.

3.5.13.2 Relay K_2.-A 1.0 volt RMS 400 cycle signal shall be injected at the commander's deflection channel ØS x sin C jack and the commander's elevation channel ØS x cos C jack. The signal shall be present at the amplifier input jacks when K_2 is de-energized. The signal shall be removed from the input jacks when K_2 is energized. When the signal is applied to IJ24-g, it shall be present at IJ26-y for both doncitions of K_2.

3.5.13.3 Relays K_3 and K_4 .- A 1.0 volt RMS, 400 cycle signal shall be injected at the commander's elevation channel parallax jack. The signal shall be present at the gunner's elevation channel parallax jack when K_1 and K_2 are de-energized. The signal shall be removed from the gunner's parallax jack when either K_4 or K_3 is energized.

3.5.14 Scale Factor Adjustments.- The scale factor adjustments of Drawing F10543800, sheets 7 and 8, shall be confirmed. The voltage ratios shall be within the ratio standard tolerances given in Table VI. Ammunition switch ISI shall be indexed to ammunitions 1, 2, 3, and 4 when Potentiometers 1R11, 12, 13, and 14 respectively are checked.

TABLE VI

POTENTIOMETER	RATIO TOLERANCE
IR15	± .000300
IR16	± .002500
IR19	± .000500

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TABLE VI (Continued)

POTENTIOMETER	RATIO TOLERANCE
IR20	± .000500
IR21	± .000350
IR22	± .000200
IR23	± .000250
IR11	± .000300
IR12	± .000300
IR13	± .000300
IR14	± .000300

3.5.15 Rate Gyro Output.- The gyro outputs given in Table VII shall be obtained for the specified rate inputs. The gyro shall be tested for CW and CCW rate inputs. A maximum transfer function coefficient shift of 0.5% is permitted after temperature cycling has been completed. The maximum time to achieve synchronous speed shall be 45.0 seconds at 115 volts ± 1% excitation.

TABLE VII

RATE INPUT	GYRO OUTPUT		
	-40°F	+77°F	+150°F
Zero rate	0.052 VRMS max.	0.025 VRMS max.	0.052 VRMS max.
4.0 mils	0.442 ± 0.068	0.450 VRMS ± 0.047	0.452 VRMS ± 0.068
12.0 mils	1.327 VRMS ±0.154	1.350 VRMS ± 0.106	1.358 VRMS ± 0.154
28.0 mils	3.096 VRMS ± 0.189	3.150 VRMS ± 0.165	3.168 VRMS ± 0.189
40.0 mils	4.423 VRMS ± 0.272	4.50 VRMS ± 0.205	4.527 VRMS ± 0.272

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwixe specified in the contract or purchase order, the supplier is responsible for theperformance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General Provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) submission.- When required by the contract, an initial production sample shall be submitted to a Government approved facility for test and evaluation in accordance with the contract schedule. The sample shall consist of three units. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection Provisions.

4.3.1 Submission of Product. - Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-D-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables VIII, IX, and X. Examination and test for Packaging and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables VIII, IX, and X shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE VIII

CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 0.65% Defective		

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TABLE VIII (Continued)

CLASS		REQUIREMENT	TEST PROCEDURE
101	Vibration	3.3.3	4.6.1.1
102	Continuity	3.4.1	Applicable Dwgs. Visual-Ohmeter
103	Input Power	3.5.1	4.6.2
104	Ammunition Selection	3.5.2	4.6.3
105	Effective Full Charge Count	3.5.3	4.6.4
106	Power Supply Output	3.5.4	4.6.5
107	Elevation Amplifier Output-Gunner's Channel	3.5.5.1	4.6.6.1
108	Elevation Amplifier Output-Commander's Channel	3.5.5.2	4.6.6.2
109	Deflection Amplifier Output-Gunner's Channel	3.5.6.1	4.6.6.3
110	Deflection Amplifier Output-Commander's Channel	3.5.6.2	4.6.6.4
111	Range Servo Amplifier Output	3.5.7	4.6.6.5
112	Parameter Resistor Package	3.5.8	4.6.7
113	Ammunition Resistor Package	3.5.9	4.6.8
114	Digital to Anal. Conv.-Relay Contacts	3.5.10.1	4.6.9.1
115	Digital to Anal. Conv.-Relay Diodes	3.5.10.2	4.6.9.2
116	Digital to Anal. Conv.- Converter Scale Factor Adj.	3.5.10.3	4.6.9.3
117	Wind X Sine Cant	3.5.11	4.6.10
118	Comm. Override Circuit	3.5.12	4.6.11
119	Boresight - Normal Circuitry	3.5.13	4.6.12
120	Scale Factor Adjustment	3.5.14	4.6.13
121	Rate Gyro Output	3.5.15	4.6.14

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TABLE VIII (Continued)

MINOR: NONE

NOTE: Tests of Table VIII shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and Rejection.-Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special Sampling.

4.4.1 General.- One unit shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table IX.

TABLE IX

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301 Fabrication	3.1	Applicable Drawing- Visual
302 General Specification	3.2	MIL-F-13926 - Visual

4.4.2 Environmental.- Three units shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table X.

TABLE X

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303 Storage temperature	3.3.1	4.6.1.2
304 Operating temperatures	3.3.2	4.6.1.3
305 Water Resistance (Sealing)	3.3.4	4.6.1.4

4.4.3 Failure of Sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.-Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment.-Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.-Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the List of Inspection Documents IEL 10543900.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2, the standard test equipment listed in Table XI shall be utilized in performing the applicable test of 4.6.

TABLE XI

	NOMENCLATURE	DESCRIPTION
1.	D.C. Power Supply (Simulated Vehicle Source)	24.0 volts ± 1%. 10 ampere output
2.	A.C. Power Supply	0 to 115 volts RMS adjustable ±1%; 400 CPS ± 1%; 2% maximum harmonic distortion. Variation in harmonic distortion shall not exceed 0.5%. Impedance less than 0.10 ohm from D.C. to 10 K.C.
3.	Oscilloscope	Sensitivity of 1 mv/cm.
4.	A.C. Ratio Standard	6 digit readout to verify ratio settings as specified in Tables III, IV, and VI. Minimum input impedance 500,000 ohms at 400 CPS, output impedance less than 10 ohms.

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TABLE XI (Continued)

NOMENCLATURE	DESCRIPTION
5. Resistance Bridge	To test resistance as specified in Tables I,II, and V.
6. A.C. Voltmeter	Reading true RMS. Accuracy of ± 1.0% of reading to measure output of 3.5.4.
7. D.C. Voltmeter	Accuracy of ± 0.1% of reading to measure output of 3.5.4.
8. Powered Rotary Rate Table	Rotate computer unit #10543800 in its operating position at adjustable angular velocities from 0.5 mil/sec to 65 mil/sec within an accuracy of 1.0% of indicated velocity.
9. A.C. Voltmeter	Accuracy of ± 0.1% of reading.

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Vibration.- To determine conformance to 3.3.3, the commander's control unit shall be mounted on a suitable mounting apparatus and attached to a vibrating machine or fixture (see 4.5.2) and vibrated at the amplitude, frequency, and time specified. After vibration, the units shall be given a visual-tactile examination for damage and loose parts and scale movement. The units shall then be subjected to a continuity test and the tests specified in 4.6.2 through 4.6.14 inclusive to determine conformance to the respective requirements referenced there in.

4.6.1.2 Storage temperatures.- To determine conformance to 3.3.1, the unit shall be tested in accordance with Temperature Cycling Procedure II of MIL-F-13926 for the specified temperatures. At the conclusion of these tests, the units shall show no evidence of damage or deterioration when visually examined.

4.6.1.3 Operating temperatures.- To determine conformance to 3.3.2, the unit shall be placed in appropriate environmental chambers (see 4.5.2). While thermally stabilized at the specified temperatures, the units shall be subjected to the tests specified in 4.6.2 to 4.6.14 inclusive to determine conformance with the requirements referenced therein.

4.6.1.4 Water resistance.- To determine conformance to 3.3.4, the unit shall be tested in accordance with Rain Procedure I of MIL-F-13926. The unit shall show no evidence of water damage as a result of this test when visually examined.

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4.6.2 Input power.- To determine conformance to 3.5.1 when units are subjected to tests 4.6.3 thru 4.6.14, the applicable test equipment items of 4.5.3 shall be checked to assure they provide proper power inputs to the terminals specified.

4.6.3 Ammunition selection.- To determine conformance to 3.5.2, the unit shall be inserted in the test circuit of Figure 4 and the control switches actuated. The corresponding indicator lights shall be observed for conformance of ammunition selection and timing.

4.6.4 Effective full charge count.- To determine conformance to 3.5.3, a single pole double throw switch shall be connected to IJ26 to simulate the EFC switch. The switch common shall be connected to IJ26-C, the normally closed contact to IJ26-B and the normally open contact connected to IJ26-A. The EFC dial in the Computer Unit shall advance 50 counts ± 1 count for each 50 actuations of the switch. A minimum of 2 seconds shall elapse between switch actuations.

4.6.5 Power supply output.- To determine conformance to 3.5.4, a load shall be connected as specified and the voltage and ripple at pin J shall be as specified when measured with the D.C.and A.C. voltmeters specified in 4.5.3.

4.6.6 Amplifier outputs.- Elevation, deflection, and range servo amplifiers of 3.5.5, 3.5.6, and 3.5.7 shall be checked by providing the specified center tapped resistive load at pin CC of IJ26 and the pins specified and monitoring the load current.

4.6.6.1 Gunner's elevation channel.- To determine conformance to 3.5.5.1, ground all test jacks except gunner's elevation jump and amplifier input. Set gain control full CCW and read voltage required at elevation jump jack to get maximum current flow through center tap.

4.6.6.2 Commander's elevation channel.- To determine conformance to 3.5.5.2, ground all test jacks except commander's elevation jump and amplifier input. Set gain control full CCW and read voltage required at elevation jump jack to get maximum current flow through center tap.

4.6.6.3 Gunner's deflection channel.- To determine conformance to 3.5.6.1, ground all test jacks except gunner's deflection and amplifier input. Set gain control to full CCW and read voltage required at deflection jump jack to get maximum current flow through center tap.

4.6.6.4 Commander's deflection channel. - To determine conformance to 3.5.6.2, ground all test jacks except commander's deflection and amplifier input. Set gain control to full CCW and read voltage required at deflection jump jack to get maximum current flow through center tap.

4.6.6.5 Range servo amplifier output.- To determine conformance to 3.5.7, ground all test jacks except range input follow up and amplifier input. Set gain control to full CCW and read voltage required at range servo follow up jack to get maximum current flow through center tap.

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4.6.7 Parameter resistor package.-To determine conformance to 3.5.8, each resistance specified in Table I shall be measured with a resistance bridge and shall be within the tolerance specified. All external power shall be removed.

4.6.8 Ammunition resistor package.- To determine conformance to 3.5.9, each resistance specified in Table II shall be measured with a resistance bridge while switch ISI is set as specified. All external power shall be remove.

4.6.9 Digital to analog converter tests.- Digital to analog converter tests of 3.5.10 shall be conducted on the following:

4.6.9.1 Relay contact.-To determine conformance to 3.5.10.1, the pins specified shall be observed for open circuit when relay K-12 is de-energized.

4.6.9.2 Relay diodes.-To determine conformance to 3.5.10.2, relay K12 shall be energized by applying 24VDC across pins IJ25-z and IJ23-B, terminal z being positive with respect to B and the voltage drop across the specified resistor shall be measured. The drop shall not be less than 5.0VDC when the resistor and 6VDC source are connected as specified.

4.6.9.3 Converter scale factor adjustment.- To determine conformance to 3.5.10.3, connect the D&A converter circuit into the test circuit of Figure 1. After determining the instrumentation ratio error with all relays de-energized, the voltage ratio for each range specified in Table III shall be measured and shall be within the tolerance specified.

4.6.10 Wind X sine cant.- To determine conformance to 3.5.11, connect demodulator and ramp generator and sine wave multiplier boards to the test circuit of Figure 2. With cant and wind input voltages as specified in Table IV, the voltage ratio shall be within the tolerance specified. The ratio of cant inputs to wind inputs shall be .666666 ± .0011 for the 6 volt input and .333333 ± .0011 for the 3 volt input during tests.

4.6.11 Commander's override circuit.- To determine conformance to 3.5.12, switch ISI shall be indexed to the ammunition no. 1 or M409 HEAT position and the resistances of Table V measured. All external power except a source to energize the relays, shall be removed. Relay K_1 shall be energized by applying 6VDC across terminals IJ24-Z and IJ26-N. Relay K_2 shall be energized by applying 21VDC across terminals IJ26-Y, terminal Y being positive with respect to B. Relay K_1 shall not be energized when K_2 is energized. Relay K_3 shall be energized by applying 21VDC across terminals IJ23-B and IJ26-S terminal S being positive with respect to B. Relay K_4 shall be energized by applying 21VDC across terminal IJ23-B and IJ26-Z, terminal Z being positive with respect to B.

4.6.12 Boresight - normal circuitry.- With switch ISI indexed to the ammunition No. 1 or HEAT M409 position, and the power sources of 3.5.1 removed, the following checks shall be made:

4.6.12.1 Relay K_1.- To determine conformance to 3.5.13.2, the signal specified shall be applied to the appropriate jacks and relay K_1 energized and de-energized. The amplifier input jacks shall be monitored for

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presence or across terminals IJ26-X and IJ23-B, terminal X being positive with respect to terminal B.

4.6.12.2 Relay K_2. To determine conformance to 3.5.13.2, the specified signal shall be applied to the appropriate jacks and the amplifier input jacks monitored. The signal shall be present at the input jacks when K_2 is de-energized and removed when K_2 is energized. When the signal is applied to IJ24-g, it shall be present at IJ26-Y for both conditions of K_2. The relay shall be energized by applying 21VDC across terminals IJ24-M and IJ23-B, terminal M being positive with respect to B.

4.6.12.3 Relays K_3 and K_4.- To determine conformance to 3.5.13.3, the specified signal shall be applied to the commander's elevation parallax jack and its presence monitored at the gunner's elevation channel parallax jack. The signal shall be present when K_3 and K_4 are de-energized. K_3 shall be energized by applying 21VDC across terminals IJ24-L and IJ23-B terminal L being positive with respect to terminal B. K_4 shall be energized by applying 21VDC across terminals IJ25-m and IJ23-B, terminal m to be positive with respect to terminal B.

4.6.13 Scale factor adjustments.- To determine conformance to 3.5.14, the computer unit shall be connected in the test circuits specified on sheets 7 and 8 of Drawing F10543800 and the potentiometer setting verified. The voltage ratios shall be within the specified tolerances. Ammunition selector switch ISI shall be indexed to ammunitions 1,2,3,4 respectively as potentiometers 1R11 through 1E14 are checked. The test circuit of Figure 4 shall be used to index ISI.

4.6.14 Rate gyro output.- To determine conformance to 3.5.15, the computer unit shall be placed on a rate table (see 4.5.3) in its vehicular position. The rate gyro shall be brought to synchronous speed in the time specified. Each rate input specified in Table VII shall be checked for gyro output voltage which shall be as specified. The A.C. voltage at pin d of IJ26 shall be adjusted to 3.000 volts ± 0.01 volts as indicated by the A.C. meter and shall remain within this tolerance while the gyro is being tested.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.-Packaging, packing, and marking shall be in accordance with MIL-P-14232 and packaging data sheet MIL-P-14232/P10543800. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Computer Unit is intended as a functional component of the electronic analoy computer, Computer Ballistics XM19.

6.2 Ordering data.- Procurement documents should specify the followings:

(a) Title, number, and date of this specification.

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(b) Selection of applicable levels of preservation, packaging, and packing.

(c) Applicable packaging data sheet number (see 5.1).

(d) Applicable stock number.

(e) Order of document precedence.

Custodian:
Army-MU

Preparing activity:
Army- MU
Project No. 1220-A263

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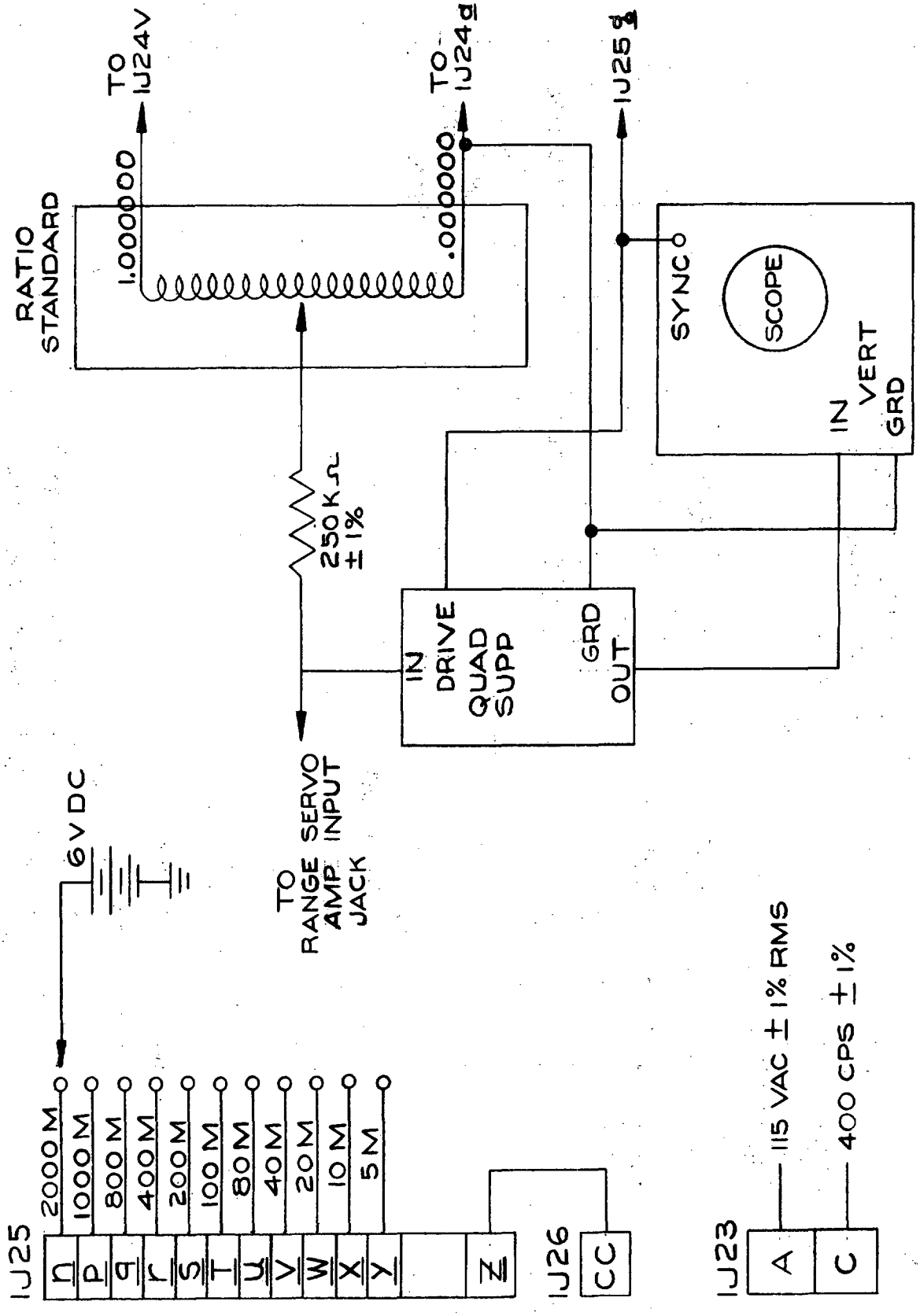


FIG. 1 DIGITAL TO ANALOG CONVERTER

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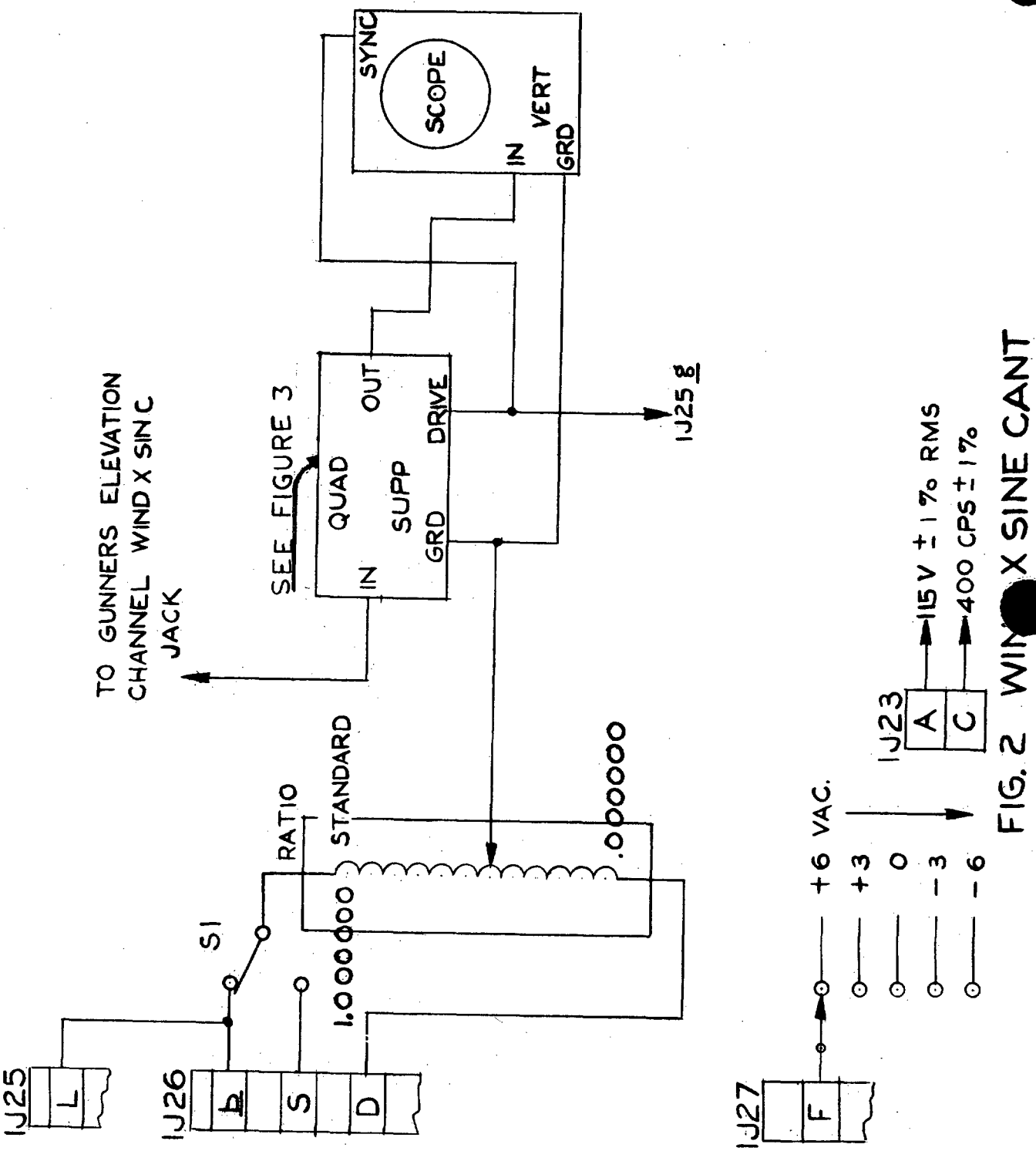


FIG. 2 WIN X SINE CANT

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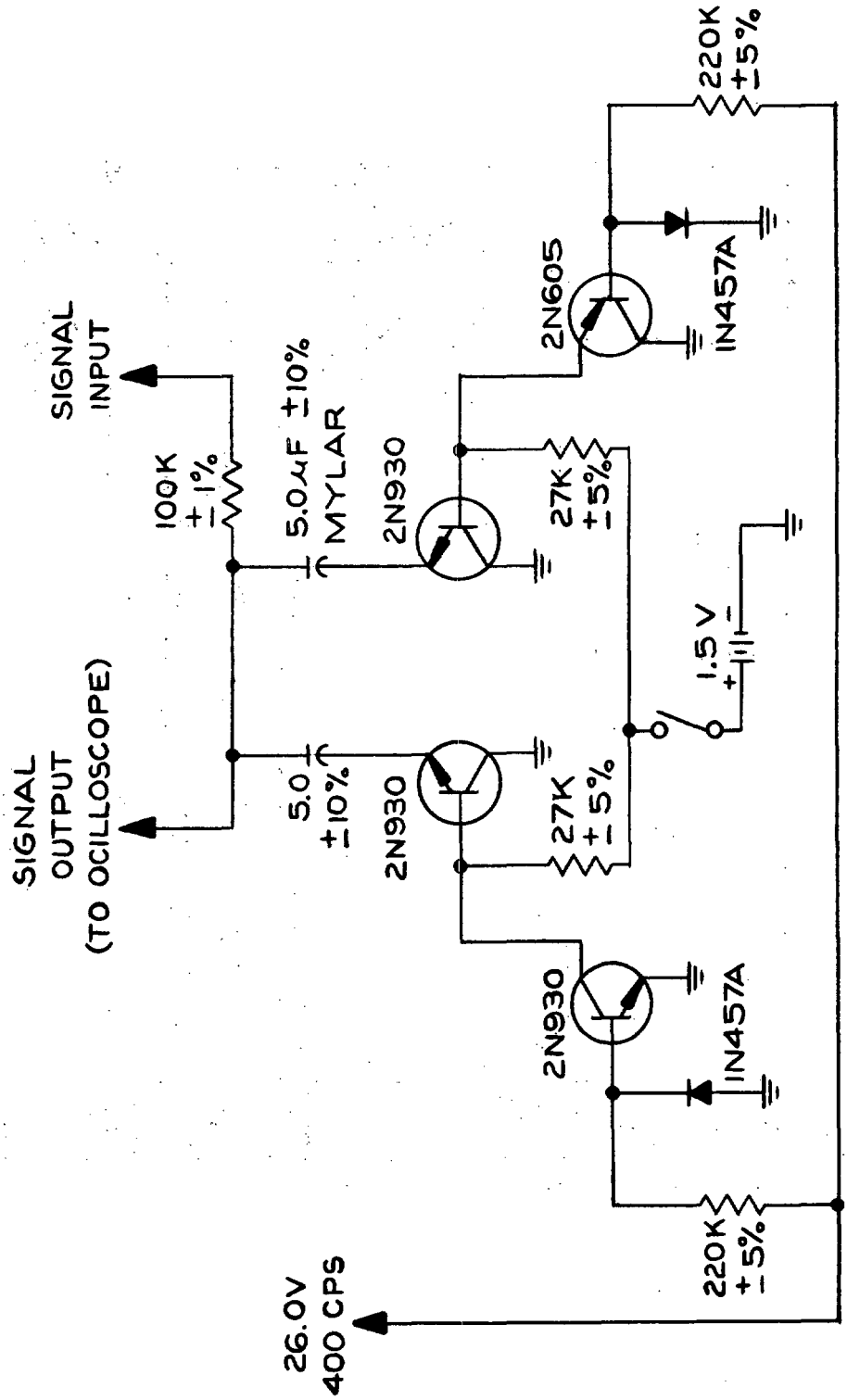


FIG. 3 QUADRATURE SUPPRESSOR

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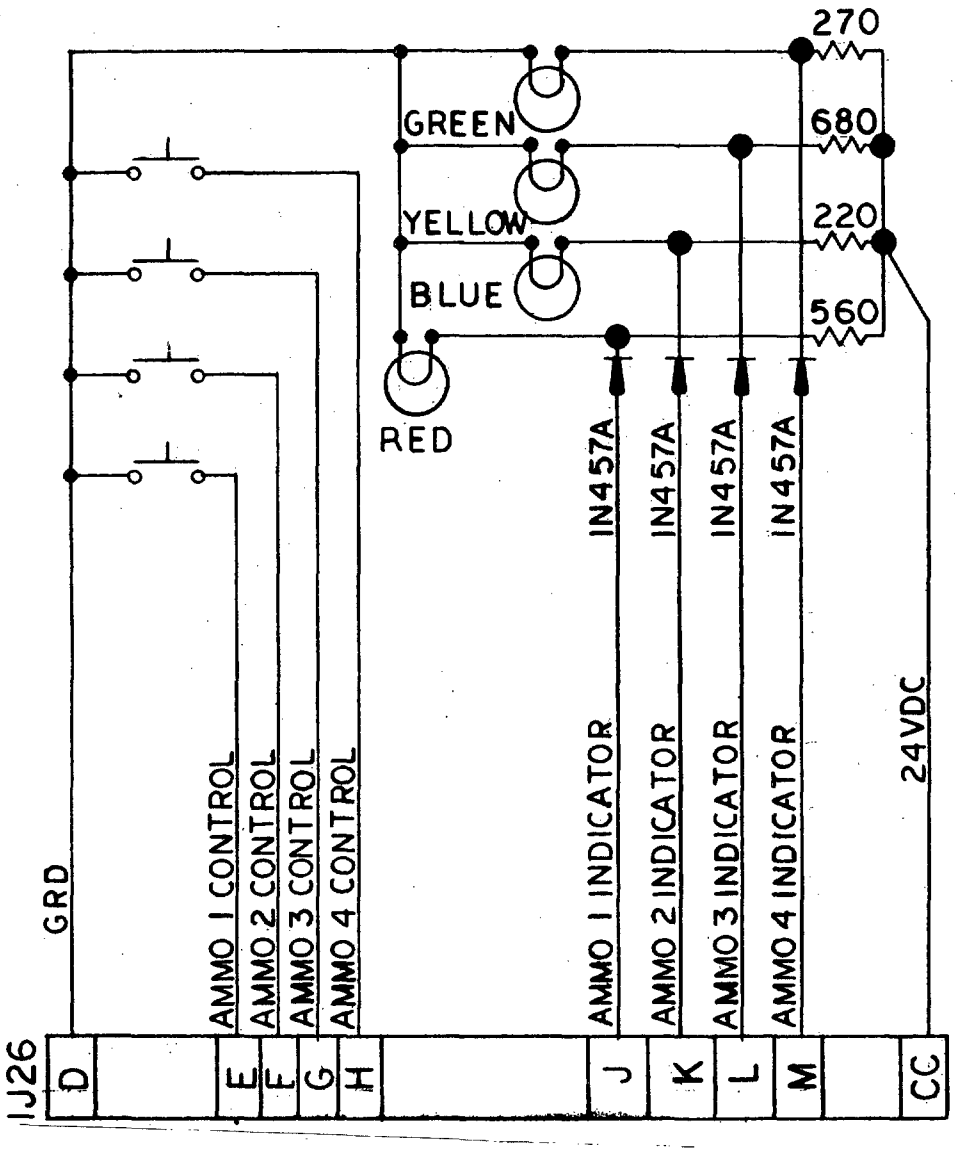


FIG. 4 AMMUNITION SELECTION

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